



10ALYTICS GLOBAL HACKATHON 2026 · TRACK B: DATA SCIENCE

SIAGA NOVA

The eye that watches five years ahead.

An early-warning intelligence layer for ASEAN public health.

POD NOVA · SDG 3 · 10 · 17 · ASEAN Digital Health & Climate Resilience Initiative

THE PROBLEM

Southeast Asia's health systems react after the crisis. By then, people have already died.

16 yrs

gap in life expectancy between the best and worst ASEAN member state

56x

higher risk of dying in childbirth in Lao PDR than in Singapore

~45%

of the global tuberculosis burden sits in the ASEAN region

The data to prevent these outcomes exists, but it is fragmented, dirty, and backward-looking. SIAGA turns it into foresight.

Three decision tools in one intelligence layer

Predict

Lifespan model

Predicts a country's life expectancy from the levers a health ministry actually controls: staffing, immunization, spend, disease burden, nutrition.

Explain

Driver isolation

SHAP ranks every driver in years of life expectancy, so policy officials see exactly where the next investment buys the most benefit.

Forecast

Disease early-warning

Backtested time-series models forecast TB and malaria five years ahead, with honest uncertainty bands.

Siaga is Malay for "alert, ready, on standby", the word ASEAN governments use for emergency preparedness.

Rigour first: we cleaned, then we proved the cleaning

18

datasets harmonized

into one clean country-year
panel: names reconciled,
headers fixed, text-numbers
parsed.

0.93

agreement with World Bank

life-expectancy correlation with
an independent source confirms
the harmonization.

8

errors adjudicated

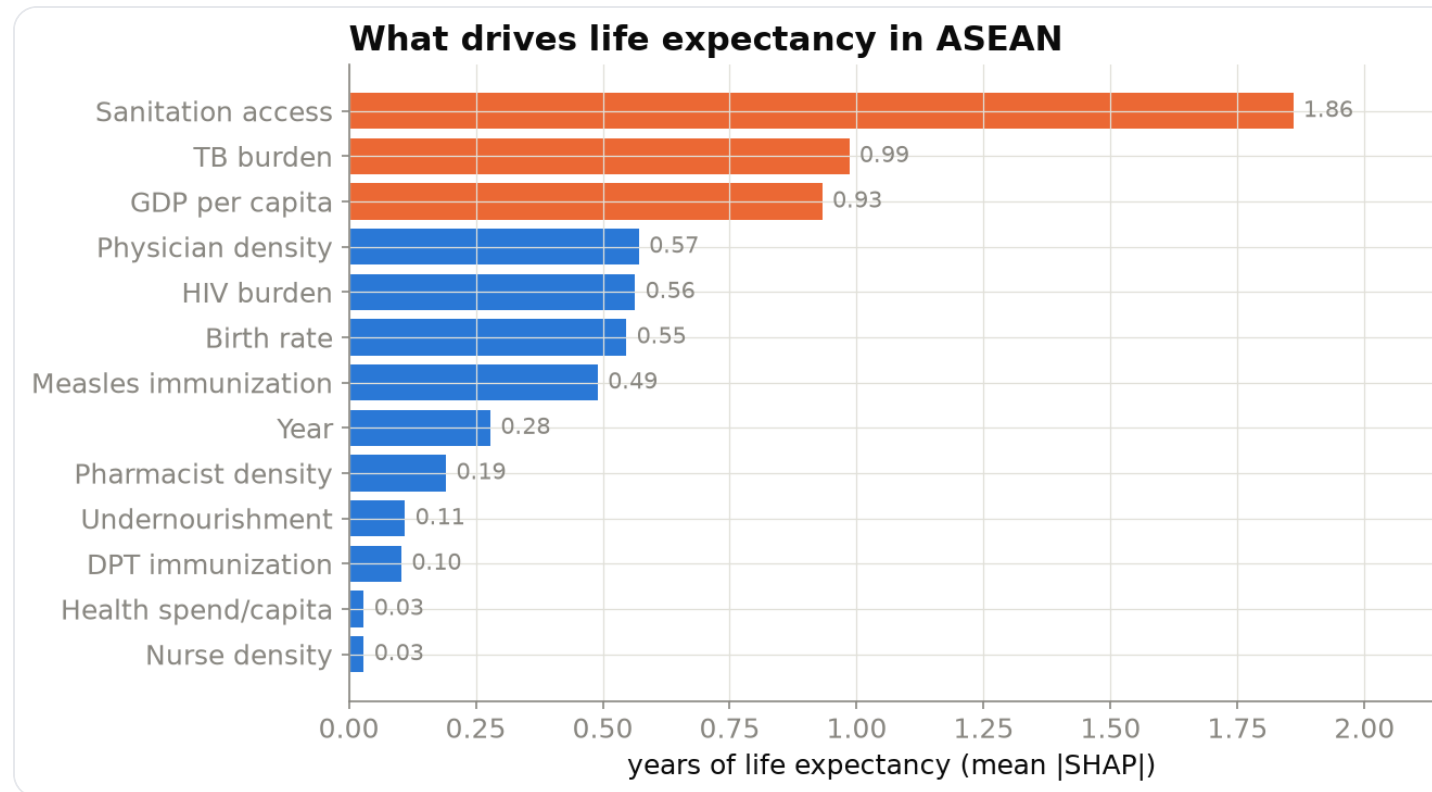
suspected entry errors
confirmed against World Bank
before repair, not dropped on a
hunch.

0

datasets discarded

every file has a role: features,
forecast targets, or the equity
layer.

Nutrition, birth rate, and disease burden move life expectancy most

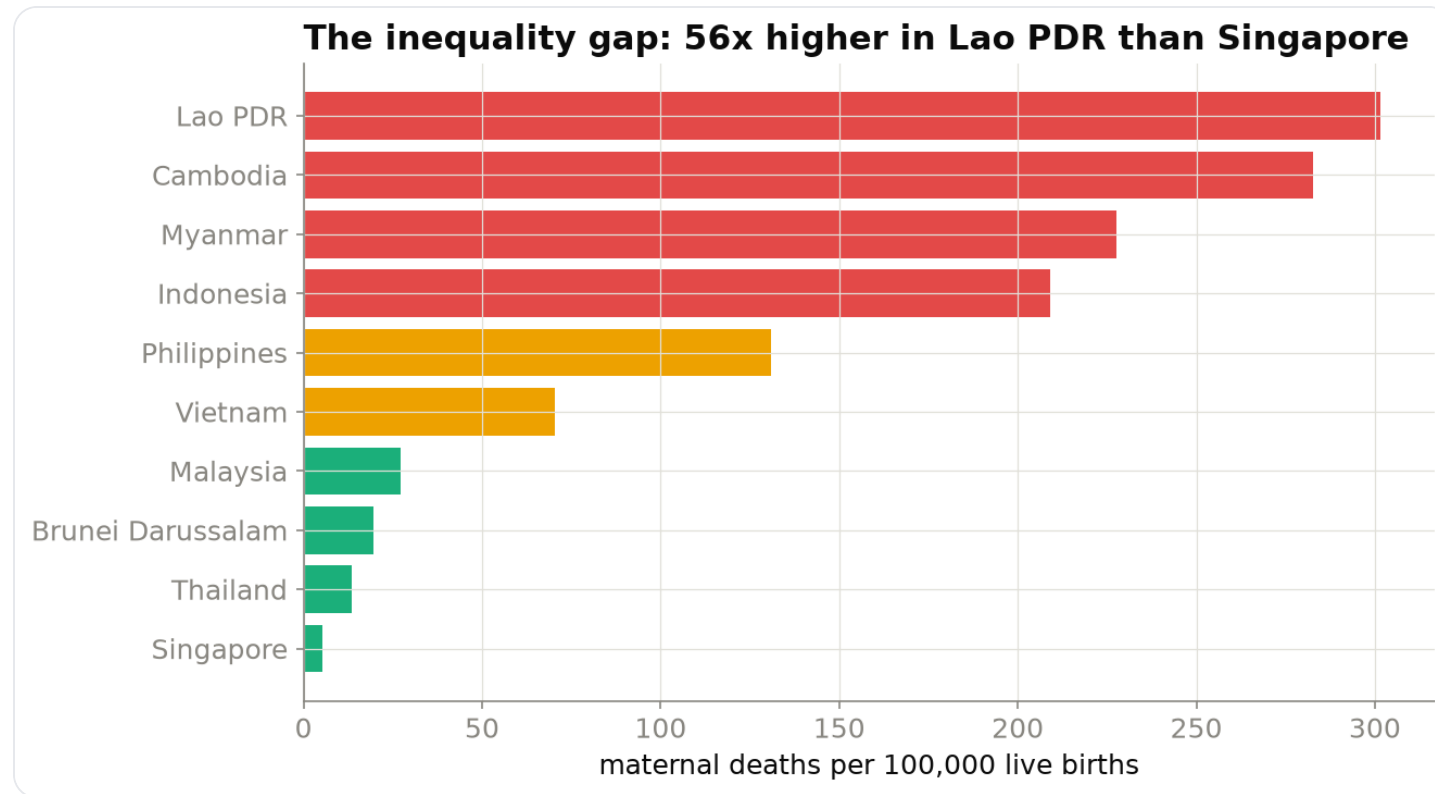


Reading it

Basic sanitation access is the single largest lever (~**1.9 years**), ahead of disease burden (TB, HIV) and income. Clean water and sanitation are historically among the strongest drivers of life expectancy.

Mortality indicators were excluded to avoid leaking the target.

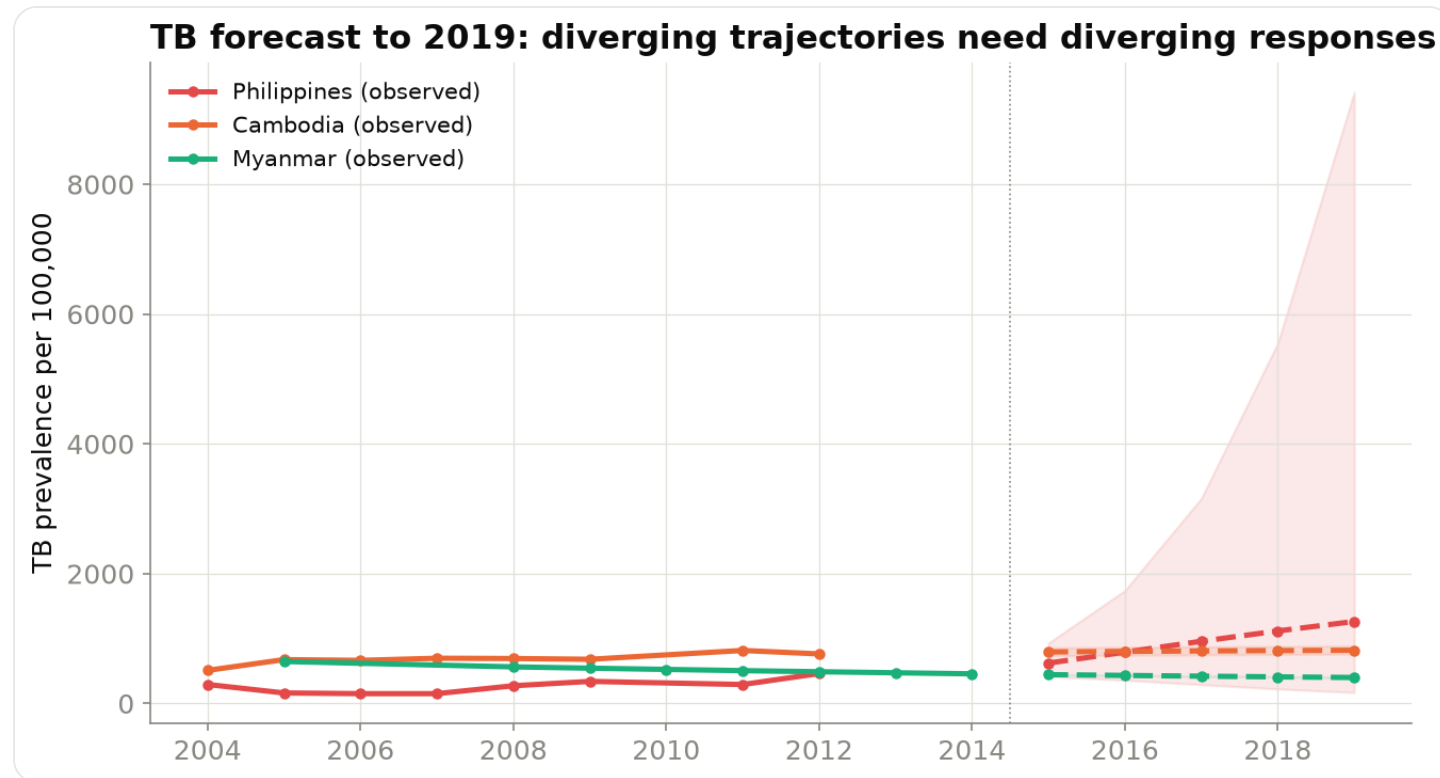
The regional average hides a 56x gap



Reading it

Maternal mortality ranges from **~5** per 100,000 in Singapore to **~300** in Lao PDR. Equity, not just averages, must guide where resources go.

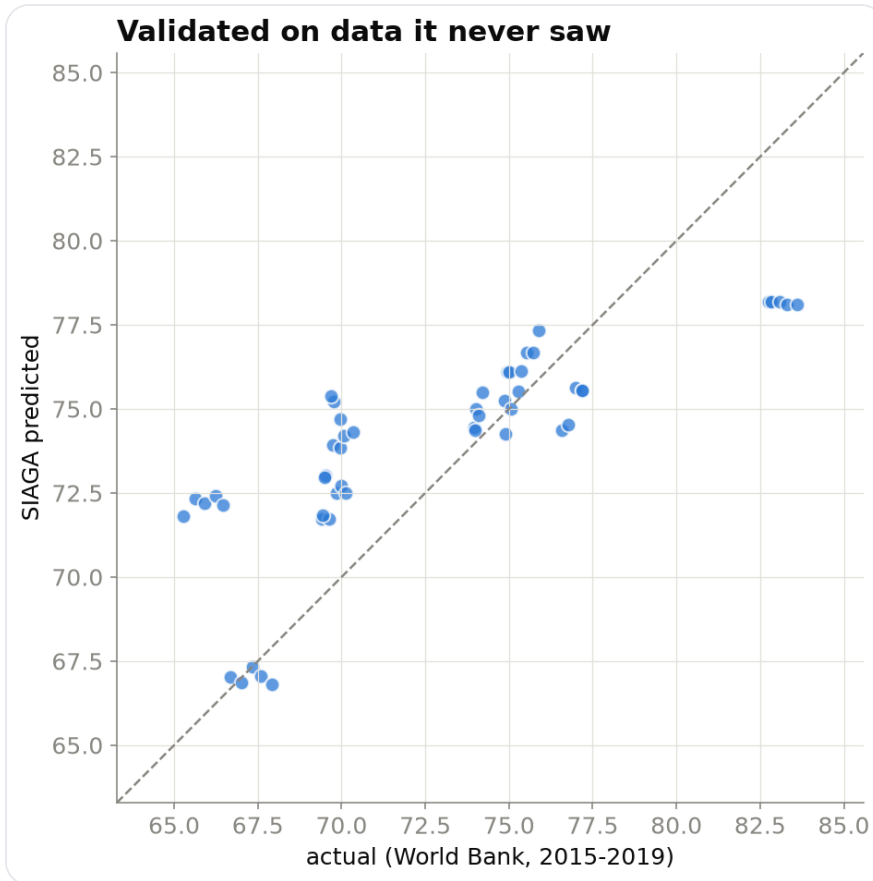
Diverging TB trajectories demand diverging responses



Reading it

Myanmar's TB is falling, but the Philippines is **rising**. A single regional policy would fail both. SIAGA forecasts each country five years out.

Honest by design, validated on data it never saw



Predict the near future of a known country

1.12 years average error · R^2 0.92 (2012 to 2014 held out)

Hold up on independent World Bank data

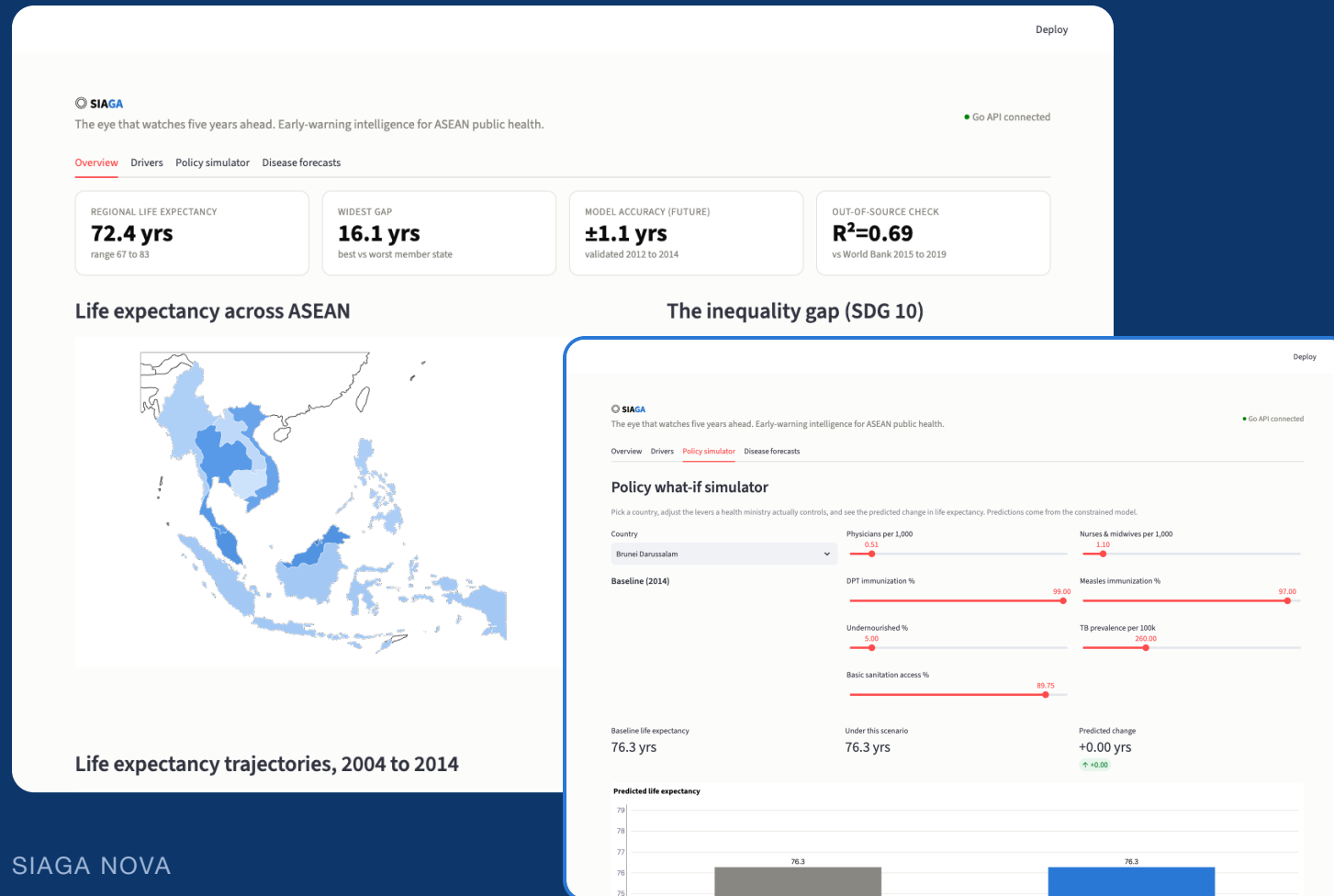
2.29 years average error · R^2 0.69, 2015 to 2019, a source the model never saw

Safe by construction

Medical priors are hard-coded, so more doctors or vaccines can never lower the prediction. The simulator cannot recommend something perverse.

THE PRODUCT

A dashboard a minister could use on Monday



Interactive policy simulator

Move a lever, see the predicted change in life expectancy, live.

Go interoperability API

A dependency-free 8 MB binary serving one harmonized schema across borders (SDG 17).
Runs on low-resource edge hardware.

Reproducible notebook

The full pipeline, from raw files to validated model, in one clean notebook.

Five actions, deployable today, grounded in the data

1

Invest in water and sanitation where coverage lags

Basic sanitation access is the model's #1 driver (~1.9 yrs) and matches the historical public-health record. Prioritise WASH infrastructure in Cambodia, Lao PDR, and Myanmar. High-return, deployable now.

2

Surge TB case-finding in the Philippines and Cambodia

Forecasts show TB high or rising. Deploy active case-finding, rapid molecular diagnostics (GeneXpert), and treatment-adherence support (WHO DOTS).

3

Fund maternal and child nutrition in high-burden states

Undernourishment remains a top driver. Scale micronutrient programmes and school feeding in Myanmar and Lao PDR, and deploy rural midwives to close the 56x maternal-mortality gap.

4

Protect immunization coverage above 95%

Measles and DPT are strong, cheap drivers. Sustain coverage and run catch-up campaigns where it has dipped.

5

Adopt one shared cross-border data schema (SDG 17)

Fragmentation delays outbreak detection. The SIAGA API lets member states share indicators for earlier regional warning.

Built to ship, designed to scale

Feasible now

Runs on free, open data and a static binary. No cloud lock-in. The API is designed for the unreliable power and connectivity the brief describes.

Real impact

Turns "report the past" into "act before the outbreak." Every recommendation ties to a measurable driver of life expectancy across 660 million people.

Phase 2 roadmap

Blockchain consent and audit layer for cross-border data. Flutter offline-first app for rural field workers. Climate and mobility data for outbreak nowcasting.

WHY SIAGA WINS

Rigorous, explainable, and honest, and it works before the outbreak, not after.



Samuel Babajide · Lead
add photo

Anne Princewill
add photo

Francis A. Mensah
add photo

Chinwem M. Onu
add photo

Christian Oamhen
add photo

David O. Agyakwa · Deputy
add photo

Pod Nova · **Vote SIAGA.**